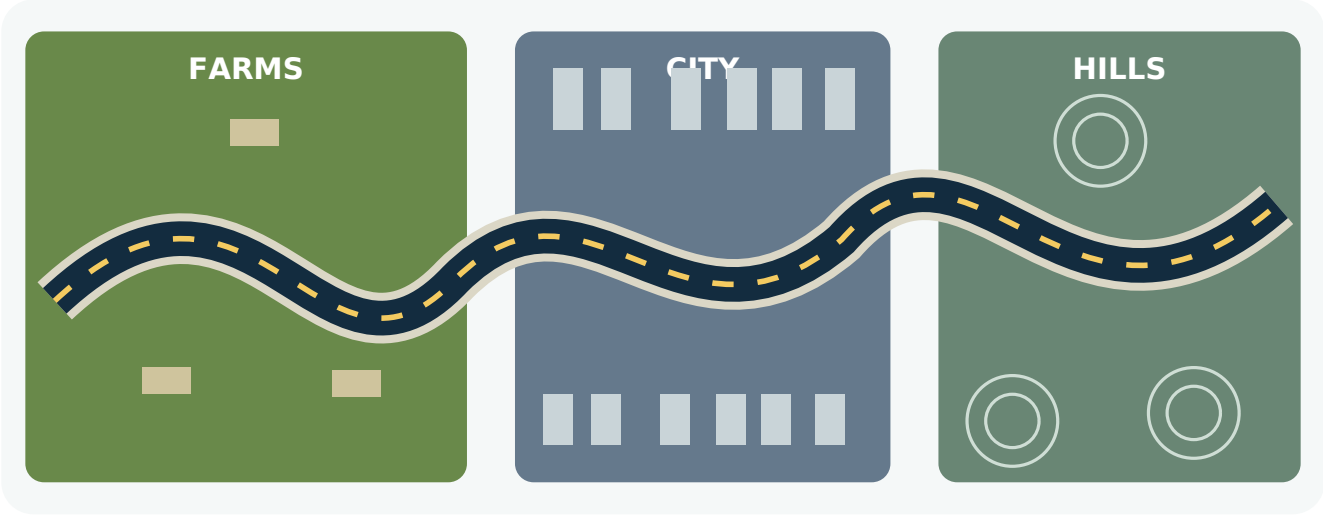
ROAD / DASH

BASE WORLD

Buildings, Terrain & Map Construction

A practical production plan for the first 6.85 km Farms → City → Hills course, excluding pedestrians, traffic, animals, combat and small props.



PROPOSED FIRST-PASS STRUCTURE TARGET

247	15	19	6,847 m
building instances	construction sections	base archetypes	existing track length

Research snapshot: RoadDash main @ 78d6a828 (2026-10-30); Road Rash native research @ 4e052aca (2026-10-30).
Counts are our new-world design targets, not the original Road Rash City asset census.

00 / How to use this specification

Decisions are distinguished from evidence. Build only structural world geometry in this pass.

Objective. Create a repeatable, physically navigable 3D level that already looks like a coherent journey even when no pedestrians, cars, cows, weapon pickups, fence posts, traffic lights, trees, or audio emitters are present.

What is in: existing point-to-point road, terrain ground surfaces, elevation and grading, roads and side-street geometry, sidewalks/curbs, subdivision parcels, 247 building instances as simple 3D volumes, structural building foundations, wall/roof massing, terrain access and collision, future interaction reservations, counts and validation tools.

What is out: pedestrians, vehicles, cattle, vegetation, signs, lamps, guardrails as detailed props, farm equipment, breakables, combat, traffic logic, detailed materials and windows, animation, music, networking, and additional courses. Empty future encounter zones may be reserved in data only.

Status language used throughout

Mark	Meaning
VERIFIED	A property observed in the current GitHub source or existing project documents.
PROPOSED	A new design decision or numerical target for this RoadDash build.
GATE	A measurable criterion the implementing agent must prove before checking it off.
DEFERRED	Intentionally excluded from structural world phase.

The single most important rule

Build the whole corridor before polishing a district

The first acceptance target is a fully continuous, drivable and walkable 6.85 km corridor with all 247 building masses spawned at valid locations. Do not spend weeks decorating 100 meters of city while the rest of the course remains empty.

Document deliverables

- 15-section count and purpose register (this PDF + machine-readable companion).
- 19-archetype structural building kit and deterministic lot placement rules.
- Road/terrain and ground-level collision requirements.
- Staged implementation tasks with acceptance tests and retrofit points in the current codebase.

01 / What the game files currently contain

Current code has working world geometry; it is not an empty project.

Current item	Verified RoadDash implementation
Course	`content/tracks/valley.json`: 14 control points, 8 grid spawns, 9 start/checkpoint/finish entries; open (not looped), about 6,847.3 meters.
Coordinate contract	`docs/tracks/AUTHORING.md`: meters, +Y up, +Z forward along spline; road half-width usually 6.5 m, separate 4.5 m safe verge.
Road geometry	`src/environment.rs`: independent rendering ribbon at approximately 1.25 m sampling and collision ribbon at 2.5 m sampling, in approximately 80 m chunks.
Ground	`src/track.rs`: analytic terrain-height sampler; `src/environment.rs`: 10 m grid terrain, colliders and road clearance.
Basic city	`src/environment.rs`: repeated boxes placed on both sides during `Sector::City`, typically height 9-34 m, windows/roofs as more cubes.
Current zoning	`rr_content::track::Sector` only defines Farms, City, Hills; no serialized district/lots/building budget yet.
Physics and tooling	Bevy 0.17.3 + Rapier 0.32.0; 60 Hz fixed gameplay; F4 track inspector and headless track validator already exist.

Sources: RoadDash Cargo.toml, src/environment.rs, src/track.rs, docs/tracks/AUTHORING.md, docs/ARCHITECTURE.md. GitHub links in the final source index.

What to KEEP

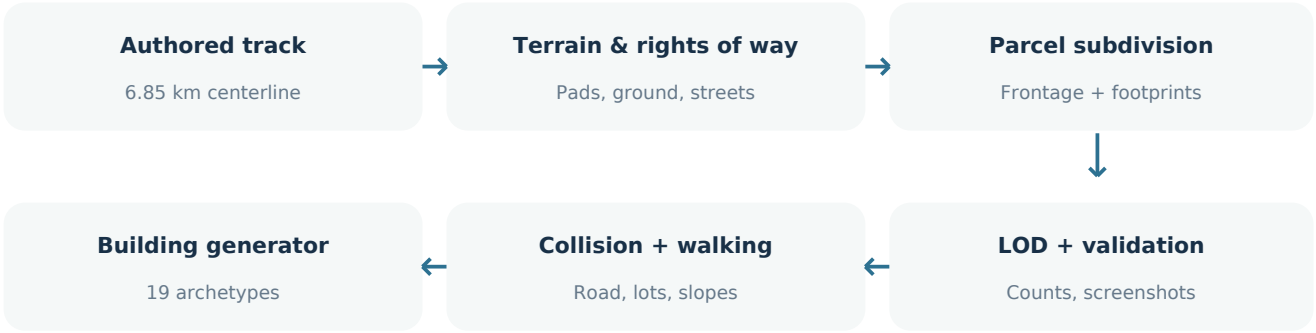
- The open road shape, checkpoint logic, fixed-timestep simulation, current road colliders and camera/recovery physics.
- The existing deterministic seed: scenery_seed = 17031; preserve seed stability or version the generation algorithm.
- The terrain-first road-clearance tests and the existing F4/F3 development visualizations.
- The physics adapter and working native Fedora/NVIDIA renderer. No Bevy/physics migration during this phase. Keep existing people, cars, animals and combat functioning; they are simply excluded from new world-construction work.

What to REPLACE incrementally

Replace repeated building spawning inside `environment.rs` with authored **district boundaries → parcel/lot subdivision → archetype selection → structural mesh generation**. Retain the original block spawner behind a debug comparison flag only during transition; avoid double-building and accidental changes to physics ownership.

02 / What the original Road Rash teaches us

Research informs the construction method, not a requirement to copy its engine.



The native C++ research shows that the original courses were stored as a **road-section graph**, with sampled curvature/elevation, lane widths, roadside ground profiles, building-family selectors, and events. Urban scenery was produced from facade, roof and exposed side panels; roads also emitted side-street and curb surfaces. That is much closer to a **road corridor generator** than a handcrafted complete downtown map.

Original data/code	Modern interpretation
NOD + HTPR	Maintain authored centerline and true 3D elevation; do not copy fixed-point projection.
NALR + DLSR	Keep varying carriageway/shoulder and add sidewalk/ROW widths as typed metadata.
DLBR terrain + building panels	Use buildings as independent lots and mesh assemblies, rather than continuous 2D facade strips.
LIHR hill profiles	Use terrain heightfield and explicit road grading, cuts/fills, and building pads.
CESR event spacing	Reserve physically valid future intersections now; postpone cars/pedestrian simulation.

The original City has 15 source course sections and a branching route, but those are not modern building counts. The original proprietary CITY.CRS/FAMILIES.RSC files are not stored in the public port, so an exact original building count is not derivable from the GitHub source alone.

RoadDash design translation

Preserve the original authored topology + controlled procedural dressing strategy. Make every roadside building a true 3D mass with an owned footprint and stable ID. Generate only enough surrounding world to support believable visibility and independent on-foot movement, not an entire open city.

03 / World scope: the 6.85 km route

The building budget is tied to the existing Valley Run geometry, not a new invented map.

Band	Approx. path distance	World identity	Terrain/building response
I. Rural	0-2.25 km	Open pasture, farm parcels	Large field polygons, sparse structures, long uninterrupted views.
II. Development edge	2.25-3.60 km	Country housing → suburb	Density rises across several sections; short cul-de-sacs and road access reserved.
III. City	3.60-4.50 km	Industrial + downtown	Continuous structural frontage; usable urban sidewalks and junction space.
IV. Hill departure	4.50-5.40 km	City fades into sloped blocks	Building density falls as terraces, cut slopes and level pads emerge.
V. Open hills	5.40-6.847 km	Ridges and rolling hills	Sparse lodges/cabins, large open terrain, deep grade changes.

The current JSON only stores Farms / City / Hills on control points. These richer five bands and 15 construction sections are PROPOSED overlays. Exact zone boundaries should be measured against the baked arc-length spline, not assumed from Z coordinates.

Spatial footprint

Recommended base-world study corridor: **0-120 m** each side of the road for authored lots and collision-relevant ground. Use **120-350 m** for background buildings, hills, and skyline blockers when visible; generated distant massing is visual-only until navigation legitimately reaches it. Keep road and walking continuity even where the ground is steep.

Terrain and street hierarchy

- Primary racing road: preserve 13 m carriageway baseline (2 × 6.5 m), with widening only where authored.
- Rural ground: open parcels separated by ground-material boundaries, not fences/props yet.
- Urban ground: curb/sidewalk and base lots, with side-road geometry reserved at junctions.
- Hills: heightfield/mesh slopes on both sides; local building pads and cut/fill, no fake flat green plane.

04 / The building number: 247 structures

This is a planning quota with explicit semantics and room for measured tuning.

247	177	70
Total building masses	Road-facing / first row	Secondary/background row

Definition: a building instance is one independently placed structural footprint, with its own persistent building ID, archetype, transform, footprint/height, collision policy, roof mass, and base elevation. A strip mall counts as **one** building even if it later has five shopfronts. A rowhouse block may count as multiple structures if each has a separate footprint and roof mass.

Not included in 247: windows, doors, roofs as separate mesh objects, fences, silos unless promoted to a counted enclosed building, sheds that are only dressing meshes, utility poles, signs, sidewalks, roads, distant billboard cards, NPCs and cars.

Category totals

Building family	Target instances	Share	Main purpose
Agricultural / rural	35	14.2%	Grounded rural approach and farms.
Residential / suburban	63	25.5%	Gradual buildup before center city.
Commercial / industrial	46	18.6%	Roadside shops, warehouses, city edge.
Urban midrise / dense	73	29.6%	City silhouette and uninterrupted street walls.
Hillside / sparse	30	12.1%	Scale and habitation without losing open hills.
TOTAL	247	100%	177 first row + 70 secondary row.

Count acceptance

- Always generate exactly 247 logical building instances for the canonical seed/version/route (unless an explicit, reviewed exclusion is approved).
- Keep every assigned structure in a validated lot; do not silently skip a building because it collides with the road. Regenerate or repair the lot and report exceptions.
- Report both logical building count and visible/loaded mesh count; these may differ under streaming or LOD.

05 / Exact section-by-section allocation

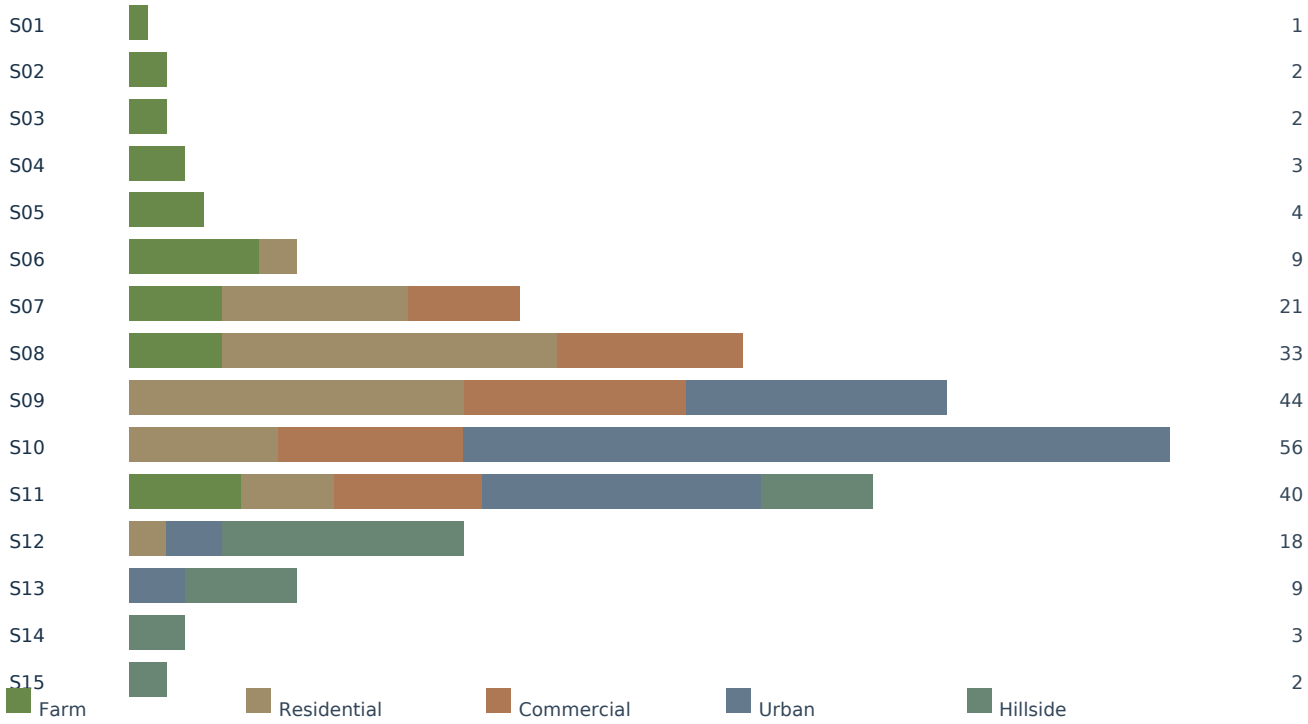
15 proposed construction sections; each covers a continuous station range.

ID / zone	Station m	1st row	2nd row	Total	Purpose
S01 Pasture Gate	0-450	1	0	**1**	Flat pasture; first sightline; minimal buildings
S02 Creek Plain	450-900	2	0	**2**	Low drainage swale and large uninterrupted ground
S03 Open Harvest	900-1,350	2	0	**2**	Fields alternate sides; no continuous frontage
S04 Long Farm Bend	1,350-1,800	3	0	**3**	Field curves, gentle grading, a farm cluster
S05 County Approach	1,800-2,250	4	0	**4**	More rural buildings; county-road threshold
S06 Rural Edge	2,250-2,700	8	1	**9**	Scattered houses among working farm parcels
S07 Edge Suburb	2,700-3,150	17	4	**21**	Houses, small commercial frontages; first side streets
S08 Residential Strip	3,150-3,600	24	9	**33**	Gridded lots and continuous sidewalks
S09 Warehouse Grid	3,600-4,050	30	14	**44**	Tighter blocks; warehouses and low-rise blocks
S10 Dense Downtown	4,050-4,500	36	20	**56**	Continuous facade rhythm; short blocks and junctions
S11 Urban Escape	4,500-4,950	26	14	**40**	Height change and mixed structures; city breaks apart
S12 Upper Estates	4,950-5,400	12	6	**18**	Hillside pads, terraced houses, road cut/fill
S13 Rolling Summit	5,400-5,850	7	2	**9**	Very sparse structures; landform dominates
S14 Ridge Descent	5,850-6,300	3	0	**3**	Large slopes, exposed land, no urban fabric
S15 Final Ridge	6,300-6,847	2	0	**2**	Finish approach; visual landscape beats structures
TOTAL	0-6,847	**177**	**70**	**247**	All sections, first-pass quotas

Distances are authoring station targets rounded from the 6,847.3 m course. Object placement positions use actual baked arc length, and section boundaries must snap to valid terrain/road joins.

06 / Visual density plan

The chart shows exactly where the building count concentrates.



Interpretation: low-density farms should feel open; built form should grow visibly over 2-3 sections before City. Dense-city structures cluster mainly in S09-S11. Once the road climbs into hills, the player should see progressively fewer buildings and larger landforms.

Building count is not the only density control

A good skyline depends on **frontage coverage, roofline variation, and lot spacing**, not only entity count. S10 is assigned 56 structures including 20 in secondary rows; those structures should not all sit in a straight line. Reserve view corridors at every junction and between key blocks.

07 / Exact building archetype library

19 structural archetypes, 247 total instances; detail meshes are deferred.

ID	Archetype	Count	Group	Height
F01	Farmhouse	10	Agricultural	6-9 m
F02	Barn	5	Agricultural	7-12 m
F03	Agricultural shed	8	Agricultural	4-7 m
F04	Rural roadside house	12	Agricultural	5-9 m
R01	Detached suburban house	28	Residential	5-9 m
R02	Townhouse/rowhouse	20	Residential	7-13 m
R03	Low-rise apartment	15	Residential	9-15 m
C01	Corner retail	11	Commercial	5-12 m
C02	Retail strip	12	Commercial	5-9 m
C03	Fuel/convenience building	3	Commercial	4-7 m
C04	Warehouse	12	Commercial	7-14 m
C05	Garage/workshop	8	Commercial	4-9 m
U01	Urban apartment midrise	25	Urban	12-30 m
U02	Office midrise	20	Urban	12-33 m
U03	Mixed-use midrise	20	Urban	12-27 m
U04	Parking structure	8	Urban	9-24 m
H01	Hillside cabin	13	Hillside	4-8 m
H02	Hillside lodge	7	Hillside	6-12 m
H03	Hill utility structure	10	Hillside	3-7 m
	TOTAL	**247**		

Mass-first generation contract

Each archetype must spawn an independent root, 1-3 primary geometric volumes, a roof cap and a simplified solid footprint. A distinct ID and a visible archetype-colored debug material are enough for the first pass. Window grids, balconies, branded signage, shutters, decorative trim, interiors and custom shaders are later passes.

Archetype heights are target ranges; the implementation should calculate floors from authored height/floor-height rules and reject footprints that cannot be supported by terrain pads.

08 / The parcel and lot generator

Make believable city blocks without hand-placing hundreds of structures.

1. Reservation pass

For every section, build a road-frame polygon containing the full asphalt/shoulder/sidewalk footprint plus design clearance. Subtract intersections, steep unsupported ground, access corridors, minimum on-foot retreat zones, route checkpoints and sightline reservations. These are non-buildable masks.

2. Frontage extraction

Extract continuous left and right road-edge polylines. Convert them into **buildable frontage spans** using exclusion intervals and corner cuts. Rural spans can exceed 100 m without any building; dense-city spans should have compact, repeating parcel widths.

3. Subdivide lots

Environment	Indicative frontage	Lot depth	Setback from sidewalk/ROW
Rural farm parcel	60-180 m	40-110 m	12-35 m
Suburban	14-28 m	25-45 m	4-10 m
Commercial strip	18-45 m	25-60 m	0-8 m
Dense city	10-24 m	20-45 m	0-3 m
Hillside residence	20-55 m	20-55 m	8-25 m

Design guidance, not zoning law or historical Road Rash dimensions. The actual front/back lot lengths must respect terrain slope and reserved lanes/sidewalks.

4. Allocate the exact quota

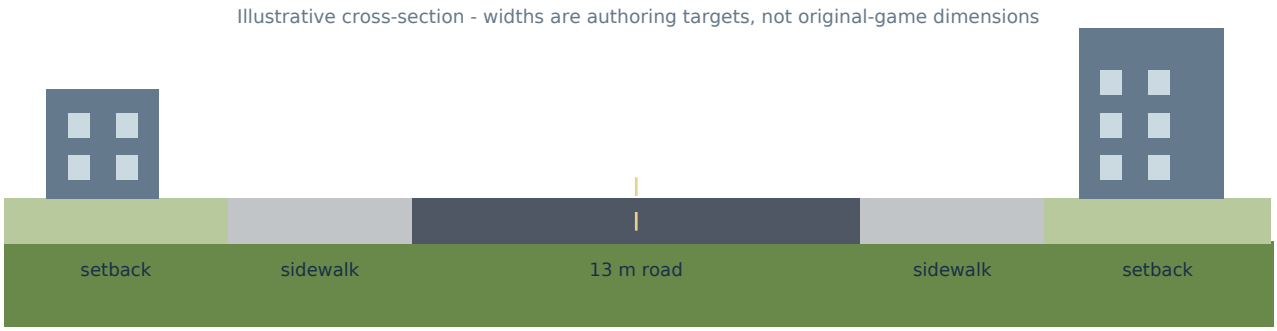
Generate an oversupply of valid candidate lots, score them by zone and frontage coverage, then select the exact count per section/group while preserving deterministic IDs. If a section cannot fit its budget, flag a failed generation gate; do not silently reduce the count or move it to another section.

5. Create the second row

Secondary structures must sit on an actual parcel grid and face a side street, local service lane, or open court; do not place them arbitrarily behind the first row. Their footprints can be simpler, but all walk-accessible ground near them must be coherent.

09 / Road and sidewalk cross-section

Ground-first geometry prevents buildings from floating and on-foot recovery from failing.



Required roadway surfaces

Surface	Generation requirement
Asphalt	Preserve existing 13 m nominal carriageway; spline tangent + bank, continuous across chunk joins.
Safe verge	Keep existing 4.5 m schema field; distinguish rideable verge from a sidewalk and from building parcels.
Sidewalk	Add urban 2.0-3.0 m broad walkable paths, lower or ramped at designated junctions.
Curbs	Create a 0.10-0.16 m vertical transition or ramp as authored; test bike and foot controller.
Side roads	Instantiate complete mesh intersections and short stub roads now, even before cross traffic exists.
Building footprint ground	Construct a level or gently terraced pad with boundary grading, not an arbitrary cube intersecting hills.

Mandatory geometry rule

Road, curb, sidewalk, lot pad and terrain must share a single height reference per world location. No visible gaps, interpenetration, terrain poking through asphalt, sharp unexplained edges, or floating/embedded building foundations.

10 / Landform generation: farms, city, hills

Terrain is an interactive 3D surface, not a colored flat base.

Farmland ground system

Generate broad field polygons and mild drainage shapes outside the road/verge corridor. Maintain open stretches between building groups. Suggested initial target: **36-48 large ground parcels** across S01-S07, with ground-color/material variation only. Do not create crop meshes, fences, cows, tractors or shrubs in this phase.

City ground system

Land surface should be graded to the road and parcel blocks, using raised sidewalk strips and building pads. Define **14-20 functional urban block faces** across the development-edge and City sections, divided by streets and buildable frontages. Count block faces and individual buildings separately.

Hill ground system

Retain the current roughly 113 m vertical elevation range, then make the landforms correspond to that path: broad convex hills, long descending slopes, occasional road cuts, hillside ledges and valleys. For structures on grades, use level pads plus short earth/retaining transitions. Do not flatten entire hills to place buildings.

Terrain area	Recommended base requirement	Acceptance
Farm fields	Large polygon patches; mild relief; open sightlines	No sudden height seams; fields do not cover road
Suburb / city	Street-level grading, smooth curb and foundation connection	Sidewalk and building bases match ground
Hills	Smooth landforms with local steeper slopes and cut/fill	Fallen rider and bike follow real terrain
Accessible space	No unintentional sealed pockets near building masses	Recovery capsule path exists back to road
Out-of-bounds	Define legitimate fall zones and rescue/return volumes	No permanent trap or accidental invisible barrier

11 / City massing rules and skyline quality

A dense city needs visible urban logic, not random cubes.

Frontage rhythm

Adjacent building faces should vary in width, roofline, frontage orientation, and setback. Preserve deliberate gaps for street openings and courtyards; avoid perfectly repeated 15 m boxes.

Block corners

Place a corner archetype at each designated intersection. Corners require visibility set-backs, curb geometry and cross-road continuity. No corner facade may sit inside a future crossing lane.

Height transitions

Near farms: 1-2 stories. Suburban: 1-4. City core: commonly 3-10. Exiting hills: progressively reduce to detached/low-rise volumes.

Roof shapes

First pass allows flat, simple pitched, and sawtooth industrial silhouettes. Rooftop water tanks, signs and vents remain decorative future props.

Back streets

Place second-row buildings on sensible internal blocks. They need street-facing orientations; keep an accessible alley or local route where the player could plausibly walk.

Visual landmarks

Author a small number of silhouette anchors: a large warehouse, a denser midrise cluster and an uphill skyline crest. These are shapes and positions, not prop-detail work.

Avoid the tempting mistake

Do not create 247 individually modeled, high-poly buildings. Spawn 247 **building instances** from 19 low-detail archetypes and a modest modular mesh library. Expand visual variety by changing floor count, footprint, footprint orientation, roof type, facade material variant and district grouping.

12 / Ground beneath each building: pads and foundations

This will save more time than any amount of decorative procedural generation.

A building must have a foundation datum derived from sampled terrain at its footprint. A rectangle spanning sloped ground cannot simply set its center to `terrain_height(x,z)` and assume all four corners are supported.

Foundation algorithm (target)

- Sample original terrain height at all footprint corners, center, and mid-edges.
- Estimate grade and cross-slope. Choose a footing strategy: level pad, stepped pad, or reject-and-relocate.
- Reserve an earthwork margin around each building (e.g. 2-5 m near urban lots, larger on hills).
- Blend the pad back into underlying terrain over an authored transition band, without raising any road/sidewalk plane.
- Generate a simplified collider for the main building and a walkable ground collider for its approach region.
- Re-sample collision terrain after earthworks; assert near-equal render and collision heights.

Recommended pad limits

District	Acceptable raw height range across footprint	Strategy
Dense city	< 0.8 m	Grade/level pad, maintain street entrances
Suburb	< 1.5 m	Pad with lawn/yard connection
Farm structures	< 2.0 m	Wide pad and earth transition
Hillside cabin	< 3.5 m	Terraced pad or stilts placeholder
Beyond tolerance	N/A	Relocate, shrink lot or explicitly author retaining structure

These tolerances are provisional world-design tunables, not building-code guidelines. Prefer less earthwork where it would distort the surrounding hills.

13 / Spatial chunking and streaming

Use current batching infrastructure, not a new world engine.

RoadDash already separates road render (~80 m chunks), terrain (~200 m chunks), and gameplay entities. Retain those independent mesh ownerships. Add **logical construction sections** of 450 m average that control content quotas, and generate **spatial runtime chunks** independently for efficient visibility.

Layer	Initial planning resolution	Owner / implementation
Macro construction	15 sections × 450 m (last 547 m)	World district catalog; quotas; debugging
Road mesh	Existing ~80 m	`src/environment.rs` road ribbon
Terrain mesh	Existing ~200 m, 10 m grid	`src/environment.rs` terrain chunks
Building logical data	One root per structure (247)	New parcel and building catalog, stable IDs
Building render mesh	Group by LOD/material and spatial chunk	Bevy mesh/material batching
Physics collisions	Simplified structural footprints in reachable corridor	Rapier fixed colliders; no separate visual physics
Distant skyline	Cheap mass-only proxy, no dynamic state	LOD/visibility pass; later optimization

World streaming requirements

- Generator must be deterministic and should not use per-frame pseudo-random placement.
- Logical world state and building IDs should remain stable even if render entities unload/reload.
- Only meshes and low-level presentation assets may downgrade with LOD; collision must remain where foot/bike interaction is possible.
- Avoid hundreds of independent window cubes; windows and detail are excluded at this stage.
- Profile 1080p native scene before adding costly graphical effects.

14 / Data schema for a reproducible world

Do not bury world-building rules in hard-coded conditionals.

Add a versioned world section file (e.g. `content/world/valley_world_v1.json`) while keeping the existing `content/tracks/valley.json` backward compatible. This structural content should deserialize and validate in `rr_content` without requiring Bevy or a GPU.

```
{
  "schema_version": 1,
  "track_id": "valley-run-01",
  "seed": 17031,
  "sections": [{
    "id": "S10", "start_m": 4050.0, "end_m": 4500.0,
    "district": "city_core",
    "roadside_buildings": 36, "secondary_buildings": 20,
    "archetype_mix": {"residential":8,"commercial":10,"urban":38},
    "lot_rules_id": "city_core_v1",
    "reserved_junctions": ["J05","J06"]
  }]
}
```

Schema validations

• Exactly 15 sections with unique stable IDs, positive lengths, no gaps or overlap, and final endpoint matching baked track length (within deliberate snapping tolerance).

• Exact quota: sum all section categories and row allocations = 247, sum near = 177, sum secondary = 70.

• All 19 archetypes referenced by a known registry; no unlicensed asset paths.

• Each building generated with stable ID, lot ID, structural class, seed, polygon, foundation datum and model/LOD references.

• Every intersection reservation mapped to a valid road station, a legal cross-road connection and minimum setback corridor.

• Generator must reject impossible parcels with a precise report, not silently skip or duplicate.

15 / World pipeline: recommended Rust boundaries

Tie this work to existing source files and avoid a wholesale refactor.

Where	Responsibility
`rr_content::world` (new)	Serialize/validate sections, district rules, archetypes and count budgets. No Bevy types.
`src/track.rs` (extend narrowly)	Read existing world coordinate system, road samples, elevation, and projection. Do not replace centerline.
`src/environment.rs` (refactor)	Keep road/terrain spawning; call dedicated world-massing subsystem instead of repeated arbitrary building spawning.
`src/world_gen/parcel.rs` (new)	Build left/right lot frontage spans, subtract reserved junctions, allocate lots.
`src/world_gen/buildings.rs` (new)	Expand exact archetype quotas to structural meshes, pads, collision, stable names.
`src/world_gen/ground.rs` (new)	Field/urban/hill pads and terrain material masks using shared height sampler.
`src/world_gen/report.rs` (new)	Deterministic generation census, validation and debug export.
`src/verification.rs` / tests	Extend smoke snapshots and no-overlap/foot-path/LOD checks; preserve existing tests.

What not to adopt yet

Do not migrate the working Bevy 0.17.3/Rapier 0.32.0 stack to newer versions merely for world generation. This phase can use current Bevy Mesh assets, existing building-cube prototype, static Rapier colliders, content serialization and F4/F3 diagnostics.

Future-proof the interface

Expose separate `WorldLayout` (stable geometry and logical building records), `WorldRender` (meshes/materials/LOD) and `WorldCollision` (structural shapes and terrain). This makes future networking and asset swaps easier without baking decisions into game logic.

16 / Implementation plan: parts B00-B10

Execute in this order. No pedestrians, cars, cattle or decorative set dressing.

B00 | Snapshot and census

Build: Read code, pin branch, export current track stats; add a visual baseline and world-counter report.

Acceptance: Verified existing track intact and current building counts captured before refactor.

B01 | Data model and quotas

Build: Create validated section/archetype/lot schemas with canonical seed and 247-building table.

Acceptance: 15 sections, 19 archetypes, all quota checks pass headlessly.

B02 | Zone station resolver

Build: Project world sections onto baked arc length; add deterministic district blending and debug gizmos.

Acceptance: No station gaps/overlaps; lookup stable at section borders.

B03 | Ground classification

Build: Add ground-material/field polygons and district terrain masks without props.

Acceptance: All 15 sections visually distinct at material/landform level.

B04 | Road + side-street foundations

Build: Keep main course road; add urban junction reservations, side-road stubs, curbs and sidewalks as structural geometry.

Acceptance: Junctions leave clear rights of way, no gaps or overlapping road surfaces.

B05 | Lot subdivision

Build: Build deterministic frontage spans, optional secondary-row plots, terrain slope rejection/repair.

Acceptance: Every parcel legal, reachable and non-overlapping.

Implementation stages — continued (B06-B10)

B06 | 247 building masses

Build: Generate one ID/root+mass/roof+collider per assigned building, with debug palette by archetype.

Acceptance: Exactly 247; 177+70; category totals; zero road/junction incursions.

B07 | Terrain foundations

Build: Grade building pads and adapt hills/terraces; synchronize physical ground and render ground.

Acceptance: No floating or embedded buildings, steep pads handled explicitly.

B08 | Walkable base world

Build: Test ground-level movement between sidewalks, accessible lots and hillside slopes after a crash.

Acceptance: No reasonable-path softlocks, no invisible walls, stable remount reach.

B09 | LOD/chunk performance

Build: Reduce mesh draw overhead and add structural LOD/visibility; retain authoritative collision.

Acceptance: 60 FPS measured target in reference native scene with representative 247-building build.

B10 | Final survey and signoff

Build: Produce screenshot atlas, GIS-like plan export, counts, source/asset provenance, regression and performance report.

Acceptance: All acceptance checklist boxes evidenced; later prop/NPC stages can start.

17 / Verification and performance gates

A completed level means repeatable geometry, safe walking and measurable performance.

Area	Required evidence
Counts	Machine-readable census with 247 IDs; 19 archetype group totals; 15 sections; 177 first row / 70 secondary.
Determinism	Regenerate twice from same seed and compare sorted (ID, archetype, transform, footprint, foundation) hashes.
Collision	Whole-footprint road/right-of-way clearance; no city/side-road blockage; pads have valid, finite support.
Terrain	Raycast along road and verge; visual and collider surfaces within tolerance; no terrain popping through asphalt.
On foot	Automated crash/walk runs in farmland, city alley, hill grade; no trapping around structural masses.
World continuity	No seams at 15 section borders, 80 m road mesh borders or 200 m terrain mesh borders.
Runtime	No per-frame building generation; bounded memory/draw calls; stable 60 Hz simulation.
Visual	Screenshot/probe at entry, each transition, dense core, uphill exit and finish approach.
Platform	Native Fedora/RTX 3080 render smoke; Windows build/CI as configured, without claiming Windows hardware unless tested.
Rights	No original EA assets, unsupported external references or third-party media included in this base implementation.

Numerical tuning gates

Start from **P95 frame <= 16.7 ms at 1080p medium** as a proposed acceptance target for the representative scene, and record entity/collider/triangle counts. A 120 FPS high-mode target can remain aspirational. Do not mark this gate passed unless native measurements are actually recorded.

For the structural phase, **1 simplified collider per near/reachable building** is the starting assumption; distant buildings can use cheap collision proxies only where navigation legitimately reaches them. Measure rather than enforcing an arbitrary maximum collider count.

18 / Gameplay-sensitive geometry, even without NPCs

The geometry must already support the future fun of crashes and cross traffic.

Cross-traffic reservations (no cars yet)

The design must include at least **7 reserved intersecting road connections** as an initial target across the rural-edge to urban/hill-departure band. Each crossing should contain a road surface, approach stub, curb/sidewalk cuts, driver sightline region, building corner setbacks, and a stable junction ID. Exact count/positions are tunable after reviewing the existing spline.

Manual movement after crashes

A crash at the edge of a block should land the rider on meaningful ground. Building colliders must not force the runner through walls or create blind dead ends. Alley entrances and accessible path widths should be reserved before detailed assets are introduced.

Hill recovery and vertical distance

At the hill crest, the ground drops away from the road. The original user goal is that a rider thrown downhill must travel actual 3D distance back to their bike, including vertical ascent. Ground generation must preserve that slope, not teleport or flatten the recovery path. Unreasonable cliff traps need a readable rescue/penalty rule.

Sightline staging

Rural sections use clear long-distance visibility. City sections create deliberate blind corners at building walls and junctions. Hill sections trade dense buildings for elevation, ridges and view occlusion. The rendering system should not spawn every building where it makes an upcoming turn or crossing unreadable.

19 / Handoff to an implementing coding agent

A precise instruction set to begin structural world work without expanding scope.

Suggested request: implement B00 through B02 first, then B03 through B06, followed by B07 through B10.

Read `AGENTS.md`, README, docs/PROGRESS.md, `docs/tracks/AUTHORING.md`, the current track JSON, `src/track.rs`, `src/environment.rs`, `src/world.rs`, and this PDF. Implement the proposed 15-section structural world without rebuilding the existing track or changing vehicle/AI/combat rules.

Introduce the exact quotas (247 buildings, 177 first-row, 70 secondary-row), make placement deterministic and data-driven, build building lots and foundations that fit the actual road and terrain, preserve future intersection right-of-way, and keep the world fully walkable. Do not place pedestrian/car/cow/vegetation/prop systems in this phase.

Session update protocol

- Only check a roadmap box when its acceptance criteria have evidence.
- If work reveals missing dependencies, add unchecked, stable-ID tasks to the correct part.
- If an additional repository is researched, log URL, commit, license and actual use in docs/third_party/REPOSITORIES.md. Do not copy unlicensed reference code.
- Keep per-section counts and generator parameters in versioned content, not hardcoded in world spawning loops.
- Run fmt, Clippy, tests, headless validation and native screenshot check where available; accurately report what was not run.
- Record screenshots at all major section boundaries and compare before/after world screenshots for regressions.

Do not do

No changes to existing NPCs, cross-traffic actors, cows, or farms as decorative props, breakable fences, street lamps, garbage cans, storefront interiors, custom motion animation, new physics engine, new game mode, or first-person walking features. This is the **terrain, roads, ground and structural building massing** pass.

20 / Source map and factual limitations

GitHub is research and code audit. Exact original assets are a separate source.

Current RoadDash source — Current playable point-to-point world, Rust + Bevy / Rapier
https://github.com/ImadMashhood/RoadDash

Valley Run track JSON — 14 control points, course sectors, spans, seed, checkpoints
https://github.com/ImadMashhood/RoadDash/blob/main/content/tracks/valley.json

RoadDash environment system — Road/terrain chunks, repeat building boxes, sidewalks
https://github.com/ImadMashhood/RoadDash/blob/main/src/environment.rs

RoadDash track/height — Arc-length spline and terrain height/clearance functions
https://github.com/ImadMashhood/RoadDash/blob/main/src/track.rs

RoadDash authoring contract — Meters, coordinate axes, track validation, graybox status
https://github.com/ImadMashhood/RoadDash/blob/main/docs/tracks/AUTHORING.md

Road Rash course decoder — Original course graph and section resource interpretation
https://github.com/arkits/road-rash/blob/main/src/course.cpp

Road Rash building renderer — Facade/roof/end faces, sidewalks, side streets
https://github.com/arkits/road-rash/blob/main/src/building_panels.cpp

Road Rash terrain decoder — Building/hill/mountain profiles
https://github.com/arkits/road-rash/blob/main/src/terrain.cpp

Road Rash course scene — Per-road scenery/terrain/material assembly
https://github.com/arkits/road-rash/blob/main/src/course_scene.cpp

Road Rash research census — Original course counts and evidence limitations
https://github.com/arkits/road-rash/blob/main/docs/research/post-11a-2026-10-12/simulation-and-courses.md

Important evidence boundary

The **247** proposed RoadDash buildings and 19 archetypes are deliberate new design choices. They are NOT claimed to be in the original Road Rash City first race. The public C++ source does not include proprietary commercial resource files, so it cannot by itself establish original building-object totals, unique facade textures or every prop identity.

The current RoadDash built-in track uses a three-sector graybox and has been progressively implemented. Counts described here are production targets, not live measurements of the current checked-in level; the implementing agent must obtain the baseline and build verification before claiming any quota is satisfied.

APPENDIX A / Implementation acceptance checklist

A finish line for this construction phase.

- ☐ Preserve and validate the 6,847.3 m point-to-point route; no accidental loop.
- ☐ Define 15 consecutive construction sections, each with ID, start/end and district.
- ☐ Introduce 19 unique structural archetypes and exactly 247 logical building records.
- ☐ Validate group totals: 35 rural, 63 residential, 46 commercial, 73 urban, 30 hillside.
- ☐ Validate row totals: 177 primary frontage, 70 secondary row.
- ☐ Generate building foundations from sampled ground; no floating or buried volumes.
- ☐ Preserve road, safe verge, kerbs, sidewalks and visible side-road foundations.
- ☐ Reserve space for at least seven proposed junction connections without placing traffic.
- ☐ Create recognizably different farmland, suburban, dense urban and hillside ground.
- ☐ Generate identical world hashes with the same seed and schema version.
- ☐ No overlap with roads, junction reservations, other lots or blocked player spawns.
- ☐ Crash recovery on foot remains navigable around urban and hill building masses.
- ☐ Road chunk and terrain chunk seams pass visual and collision checks.
- ☐ No decorative small props, NPCs, animals, traffic or copyrighted game assets added.
- ☐ Collect and review start/farm/city/crest/finish screenshots and native performance.
- ☐ Update progress, decisions, third-party source log, tests and known issues.

Signoff condition

The base-world pass is ready for the next environment phase when all 247 building masses are legally and physically placed, road/ground continuity and collision are verified, all 15 sections read as distinct locations, and the existing motorcycle gameplay/regression gates still pass.

APPENDIX B / Section and archetype number audit

These arithmetic identities must remain true when tunables change.

Quantity	Required total	How calculated
Building instances	247	Sum(section.near + section.secondary)
Near road	177	Sum(section.near)
Secondary row	70	Sum(section.secondary)
Agricultural	35	Sum(section.farm) = sum(F01-F04)
Residential	63	Sum(section.residential) = sum(R01-R03)
Commercial	46	Sum(section.commercial) = sum(C01-C05)
Urban	73	Sum(section.urban) = sum(U01-U04)
Hillside	30	Sum(section.hill) = sum(H01-H03)
Construction sections	15	S01-S15 cover 0 .. 6,847.3 m
Building archetypes	19	4 + 3 + 5 + 4 + 3 classes

Suggested phase-1 output files

- `content/world/valley_world_v1.json` - station sections and quotas
- `content/world/building_archetypes_v1.json` - archetype dimensions/rules
- `src/world_gen/parcel.rs` - buildable lots
- `src/world_gen/buildings.rs` - structural masses and colliders
- `src/world_gen/report.rs` - JSON/CSV world census
- `docs/world/BASE_WORLD_TESTS.md` - screenshots, metrics, validation evidence

A ready-to-edit section CSV and draft JSON specification are delivered alongside this PDF. These are proposed content-design inputs; they are not yet merged into your GitHub repository.